

Diagnosing Effective Latent Dimensionality in Autoencoder Representations via Active-Unit Dynamics and Intrinsic-Dimension Estimates

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Abstract

Determining how many latent coordinates an autoencoder actually uses is difficult because reconstruction error, intrinsic-dimension estimates, and variational activity measure different aspects of the learned representation. This paper proposes a multi-indicator diagnostic framework for interpreting effective latent dimensionality in autoencoders (AEs) and variational autoencoders (VAEs). The framework combines active-unit (AU) dynamics, TwoNN/MLE intrinsic-dimension (ID) estimates, reconstruction-error elbows, and PCA baselines. It does not treat the AU slowdown point, m^{knee} , as a unique optimal bottleneck size. Instead, AU counts serve as direct observations of coordinate usage in VAEs; TwoNN/MLE estimates as numerically stable auxiliary references subject to geometric bias; and m^{knee} as an exploratory, grid- and seed-dependent change point.

Under a linear-Gaussian β -VAE rate-distortion approximation, we derive the activation condition $\lambda_j > \beta\tau^2$, which provides an interpretable water-filling view of KL-induced dimension suppression. Experiments on fully connected VAEs trained on MNIST show that AU counts and TwoNN estimates are stable across random seeds, whereas sparse-grid m^{knee} is unreliable (43 ± 15). A dense grid around the estimated intrinsic dimension stabilizes m^{knee} at 10.3 ± 0.9 , consistent with TwoNN near 10. Additional experiments on Conv-VAEs, CIFAR-10, and SVHN clarify boundary conditions under which $\text{AU}/m^{\text{knee}}$ identification fails; Fashion-MNIST is used as a preliminary external check only. These results support the proposed framework as a diagnostic protocol for latent-dimensionality analysis rather than a universal dimension estimator.

Keywords: variational autoencoder; effective latent dimensionality; active units; intrinsic dimension; TwoNN; diagnostic framework; representation learning

1 Introduction

1.1 Background

Autoencoders and variational autoencoders (VAEs) [13] learn compact representations of high-dimensional data by forcing information through a bottleneck of dimension m . A key design question is: *how many of the m latent coordinates does the model actually use?* This “effective latent dimensionality” governs reconstruction fidelity, disentanglement, and posterior collapse [22].

In the context of machine learning and knowledge extraction, latent representations underpin model selection, compression, downstream prediction, clustering, and visualisation [8]. However, the nominal bottleneck dimension m does not necessarily coincide with the number of coordinates the model actually exploits. A diagnostic protocol that distinguishes coordinate usage, geometric intrinsic dimension, and reconstruction-side behaviour is therefore directly useful for interpreting learned representations and selecting latent-space sizes in a reproducible manner.

Existing work measures individual aspects of latent geometry—AU counts [22, 21], ID estimates [9, 10], isometric regularisation [14]—but a systematic framework that combines these indicators, clarifies their distinct roles, and defines the conditions under which each is reliable has not been established.

1.2 Problem Statement and Contributions

The central research question is:

How can we diagnose, without over-claiming, how many dimensions an AE/VAE effectively exploits, and where does such diagnosis break down?

The contributions of this paper are threefold.

1. **Multi-indicator diagnostic framework.** We propose a framework that explicitly assigns distinct roles to four indicator classes: (i) AU values as direct usage observations; (ii) TwoNN/MLE ID as numerically stable auxiliary references subject to geometric bias; (iii) m^{knee} as an exploratory change point; and (iv) MSE elbow and PCA as reconstruction-side baselines. The main contribution is not m^{knee} itself but the role separation, which prevents users from over-relying on any single indicator. MNIST three-seed experiments confirm that AU values ($\sigma \leq 0.5$) and TwoNN ID (10.03 ± 0.12) are reproducible; the main stabilised result for m^{knee} is 10.3 ± 0.9 under dense-grid, independent-initialisation, multi-seed conditions.
2. **Linear-Gaussian rate-distortion analysis.** Under a linear-Gaussian β -VAE rate-distortion approximation, we derive the activation condition $\lambda_j > \beta\tau^2$ and show that AU saturation follows a water-filling principle. This analytical result is an *interpretive approximation model* for the nonlinear setting, not a general theorem for arbitrary VAEs.
3. **Empirical characterisation of reliability and limits.** Dense-grid/multi-seed experiments show that sparse-grid m^{knee} is unreliable ($\sigma = 15$), whereas dense-grid evaluation stabilises it near the TwoNN estimate. Conv-VAE and natural-image experiments identify boundary conditions under which AU/ m^{knee} diagnostics fail or require stronger regularisation and architectural control.

1.3 Paper Organisation

Section 2 reviews related work. Section 3 defines the diagnostic framework. Section 4 provides theoretical background. Section 5 reports experiments. Section 6 discusses findings. Section 7 concludes. Detailed tables and proofs are in the Appendix.

2 Related Work

Representation learning and autoencoders. Bengio et al. [8] provide a comprehensive review of representation learning, including the manifold hypothesis [7] underpinning AEs. β -VAE [21] increases KL weight to encourage disentanglement; Lucas et al. [22] analyse posterior collapse in the linear-VAE limit. Contractive AEs [23] and denoising AEs [24] offer complementary regularisation strategies.

Intrinsic-dimension estimation. TwoNN [9] and MLE [10] estimate local intrinsic dimension from nearest-neighbour distances. Pope et al. [17] apply TwoNN to natural images, reporting $d_{\text{ID}} \approx 26$ for CIFAR-10 at 32×32 . Ansuini et al. [18] track ID across deep-network layers; Valeriani et al. [20] extend this to large transformer models.

Geometric autoencoders. Nazari et al. [15] and Kato et al. [14] propose isometric regularisation of the decoder Jacobian. We use IsometricAE as a controlled testbed for geometric bias, not as the main tool.

Positioning. The present work differs from the above in that it targets the *design stage*—systematically studying how AU dynamics and ID estimation jointly characterise effective dimensionality, and quantifying when each indicator is reliable.

3 Diagnostic Framework

3.1 Problem Setup

Let an AE/VAE have encoder $f : \mathbb{R}^N \rightarrow \mathbb{R}^m$ and decoder $g : \mathbb{R}^m \rightarrow \mathbb{R}^N$. For a VAE, the encoder outputs a posterior $q_\phi(z|x) = \mathcal{N}(\mu(x), \sigma^2(x))$; KL regularisation (β -weighted) acts as an information bottleneck.

Definition 1 (Active Units [22]).

$$\text{AU}(m) = \sum_{k=1}^m \mathbf{1}[\text{Var}_x[\mu_k(x)] > \delta], \quad \delta = 10^{-2} \quad (1)$$

where $\mu_k(x)$ is the k -th posterior mean. AU is robust to the choice of $\delta \in [10^{-3}, 10^{-1}]$ (Appendix E.1).

Definition 2 (AU Slowdown Point). Let $s(m_i) = (\text{AU}(m_i) - \text{AU}(m_{i-1})) / (m_i - m_{i-1})$ be the raw AU increase rate, and $\rho(m_i) = s(m_i) / \max_j s(m_j) \in [0, 1]$ the normalised rate. The slowdown point is

$$m^{\text{knee}} = \min\{m_i \in \mathcal{M} : \rho(m_i) < \tau_{\text{knee}}\}, \quad \tau_{\text{knee}} = 0.25. \quad (2)$$

Role separation. Table 1 summarises the roles.

Table 1: Classification and empirical reliability of diagnostic indicators. *Robust* means low inter-seed variance under tested conditions, not unbiasedness toward the true ID. *Exploratory* means useful for hypothesis generation but not a unique estimate of effective dimensionality.

Indicator	Category	Empirical reliability	Primary use
AU values	Stable across seeds [‡]	$\sigma \leq 0.5$ (MNIST 3-seed)	Direct observation of active coords (not unbiased toward true ID)
TwoNN/MLE ID	Numerically stable auxiliary [†]	$\sigma \leq 0.12$ ($m=64$)	Reference ID estimate (stable, not unbiased)
MSE elbow	Auxiliary	Qualitative only	Reconstruction-side check
PCA cum. variance m^{knee}	Auxiliary Exploratory	Linear assumption $\sigma = 15$ sparse; 0.9 dense	Linear baseline Rough diagnostic heuristic

[†]Subject to isometry bias; numerically stable but not unbiased toward true d_{ID} . [‡]“Stable across seeds” means low inter-seed variance under tested conditions, not unbiasedness. m^{knee} is not a standalone estimator; informative only under dense-grid, independent-initialisation, multi-seed evaluation.

3.2 Two-Stage Diagnostic Pipeline

Stage 1 (ID estimation, auxiliary). For synthetic data with known d_{true} , TwoNN/MLE is applied directly. For real data (e.g. MNIST), a large reference AE ($m_{\text{ref}} = 256$) is trained; its latent representation is fed to TwoNN/MLE to obtain $\hat{d}_{\text{ID}}$. *Limitation:* this reference AE is not isometric ($\kappa \approx 10^9$); $\hat{d}_{\text{ID}}$ is an approximation subject to geometric bias.

Stage 2 (AU Dynamics, primary). A VAE ($\beta = 4$) is trained on a grid $\mathcal{M} = \{1, 2, 4, 8, 12, 16, 20, 32, 64, 128\}$. For *sharp-saturation* data (synthetic manifolds), AU_{sat} is read directly. For *gradual-increase* data (real data), m^{knee} is computed from Eq. (2) and cross-checked against $\hat{d}_{\text{ID}}$ from Stage 1.

4 Theoretical Background

4.1 Linear-Gaussian Rate-Distortion Analysis

Analytical Result 1 (AU Activation Condition in Linear-Gaussian Rate-Distortion Surrogate). *Consider a β -VAE with linear encoder/decoder and Gaussian prior, analysed under a linear-Gaussian rate-distortion surrogate approximation (details in Appendix B.2). Under this approximation, the water-filling condition implies:*

1. (**Activation threshold**) *The j -th eigen-direction becomes active iff $\lambda_j > \beta\tau^2$, where λ_j is the j -th data-covariance eigenvalue, τ^2 is the decoder noise variance.*
2. (**Upper bound**) *The optimal number of active eigen-directions satisfies $A^*(m) \leq \min(m, d^*(\beta))$, where $d^*(\beta) := |\{j : \lambda_j > \beta\tau^2\}|$.*
3. (**Monotone saturation of optimal solution**) *$A^*(m)$ is non-decreasing in m and saturates at $d^*(\beta)$ for $m \geq d^*(\beta)$.*

Remark 1 (Empirical AU is non-monotone). *Analytical Result 1(3) concerns $A^*(m)$, the global-optimum active eigen-direction count. In practice, each bottleneck size m is trained independently*

with stochastic optimisation; the resulting empirical $AU(m)$ can be non-monotone due to finite-sample effects and initialisation dependence (e.g. $AU=29$ at $m=64$ but $AU=26$ at $m=128$, Table 8). Therefore, m^{knee} should be treated as an exploratory diagnostic indicator, not a precise estimate of $d^*(\beta)$.

Remark 2 (Vanilla AE corresponds to $\beta = 0$). Without KL regularisation, the threshold $\beta\tau^2$ vanishes, so all directions with positive variance are active. Thus the pseudo-AU (coordinate-usage count) of a standard AE approximates m .

4.2 Grid-Resolution Dependence of m^{knee}

Proposition 1 (Grid Resolution Dependence). Under an idealised monotone-saturation model where $AU(m)$ changes like a step function at $d^*(\beta)$ without noise, the detection error satisfies $|m^{\text{knee}}(\mathcal{G}) - d^*(\beta)| \leq \max_i(m_{i+1} - m_i)$.

Remark 3. This bound applies to the idealised model, not real non-linear VAEs whose AU curves fluctuate non-monotonically. The practical implication is qualitative: finer grids near the estimated $\hat{d}_{\text{ID}}$ reduce the variance of m^{knee} , as confirmed in Section 5.4.

4.3 Pullback Metric and Isometry

Proposition 2 (Pullback Metric and Local Isometry). The decoder $g : \mathbb{R}^m \rightarrow \mathbb{R}^N$ is locally isometric at z iff $G(z) = J_g(z)^\top J_g(z) = I_m$ (all singular values equal 1). The condition number $\kappa = \sigma_{\max}/\sigma_{\min}$ measures anisotropy: $\kappa = 1$ means all singular values are equal but not necessarily 1. Therefore κ is an anisotropy indicator, not a sufficient condition for isometry.

4.4 Design Principle: Bottleneck Margin and TwoNN Stability

Experiments in Section 5.5 support the following design principle.

Design Principle 1. The dominant factor for numerical stability of TwoNN estimates is a sufficient bottleneck margin ($m \gg \hat{d}_{\text{ID}}$), not isometric regularisation. TwoNN underestimates when $m \approx d_{\text{true}}$; it converges once $m \geq 5\hat{d}_{\text{ID}}$ approximately.

This is an empirically supported guideline (verified on MNIST and Fashion-MNIST across four model types) rather than a formal theorem.

5 Experiments

5.1 Experimental Setup

All models use Adam ($\text{lr} = 10^{-3}$), batch size 128, ReLU activations, Sigmoid output, and PyTorch 2.0.

Synthetic manifolds. Hidden layers [64, 32], 200–300 epochs, $\beta = 4.0$, $n = 5000$.

MNIST / Fashion-MNIST. Input $N = 784$, hidden [512, 256], $\beta = 4.0$, grid $\mathcal{M} = \{1, 2, 4, 8, 12, 16, 20, 32, 64, 128, 256\}$. MNIST: 300 epochs, $n_{\text{train}} = 20\,000$, $n_{\text{test}} = 3\,000$. Fashion-MNIST: 100 epochs, $n_{\text{train}} = 10\,000$.

ID estimation: TwoNN ($k = 2$), MLE ($k = 10$), AU threshold $\delta = 10^{-2}$. Three random seeds $\{42, 123, 777\}$ for main reproducibility experiments. All experiments in the multi-seed study use independent per- (m, seed) initialisation.

5.2 Synthetic Manifolds: Principle Verification

5.2.1 MSE Elbow and Manifold Dimension

On Swiss Roll ($d_{\text{true}} = 2$, simply connected) and Torus ($d_{\text{true}} = 2$, $d_{\text{emb}} = 3$), the MSE of a standard AE shows a sharp elbow at $m = d_{\text{true}} = 2$ (Swiss Roll) and $m = d_{\text{emb}} = 3$ (Torus). This is explained by the topological embedding dimension: although both manifolds have intrinsic dimension 2, the Torus cannot be embedded in $\mathbb{R}^2$ without self-intersections, whereas a standard smooth embedding requires $\mathbb{R}^3$. The reconstruction lower bound therefore depends on d_{emb} , not only d_{ID} (Appendix B).

Table 2: MSE of AE on synthetic manifolds (mean $\pm$ std, 3 seeds). Swiss Roll elbow at $m = d_{\text{true}} = 2$; Torus elbow at $m = d_{\text{emb}} = 3$. The rise at $m = 4$ for Swiss Roll (std= 0.015, vs. 0.002 at $m = 2$) reflects optimisation variance when the extra dimension is unused: higher std indicates sensitivity to random initialisation.

m	Swiss Roll MSE	Torus MSE	Note
1	1.016 ± 0.057	0.388 ± 0.017	under-parameterised
2	0.029 ± 0.002	0.085 ± 0.004	elbow (Swiss Roll)
3	0.0008 ± 0.0001	0.0005 ± 0.0000	elbow (Torus)
4	0.041 ± 0.015	0.019 ± 0.002	high std: optimisation variance

5.2.2 AU Saturation on Synthetic Data

VAE ($\beta = 4$) saturates at $\text{AU}_{\text{sat}} = 2$ (Swiss Roll, $= d_{\text{true}}$) and $\text{AU}_{\text{sat}} = 3$ (Torus, $= d_{\text{emb}} > d_{\text{true}}$), both with zero inter-seed variance, consistent with Analytical Result 1 (qualitative). For standard AE, pseudo-AU= m (all coordinates active), consistent with the $\beta = 0$ remark of Analytical Result 1. Detailed topological-manifold results (Torus, Möbius Band) are in Appendix C.

5.3 MNIST: Multi-seed Reproducibility Study

5.3.1 ID Estimation via Reference AE

A standard AE trained on MNIST (300 epochs) with $m = 256$ yields TwoNN ID ≈ 10.02 , MLE ID ≈ 11.85 . Multiple m values confirm convergence above $m \geq 64$: TwoNN ID= 9.54 at $m = 64$, stabilising to 10.02 at $m = 256$. This places MNIST’s intrinsic dimension at $\hat{d}_{\text{ID}} \approx 10\text{--}12$ (Table 3).

5.3.2 VAE AU Dynamics: Gradual Growth Observed

MNIST VAE ($\beta = 4$) shows gradual, non-monotone AU growth as m increases, consistent with non-linear stochastic optimisation (e.g., AU= 29 at $m = 64$ but AU= 26 at $m = 128$). This gradual growth is the key reason why sparse-grid m^{knee} is unreliable: the AU curve does not have the sharp step assumed in the idealised model (Proposition ??). A single-seed preliminary run

Table 3: MNIST AE results (300 epochs, seed 42).

m	MSE	TwoNN ID	MLE ID
1	0.0437	1.02	1.15
32	0.0087	8.84	9.89
64	0.0062	9.54	11.26
256	0.0057	10.02	11.85

illustrating this pattern is provided in Appendix A (Table 8). **The $m^{\text{knee}} = 12$ value observed in that run is not used as a main result**; the main stabilised result is $m^{\text{knee}} = 10.3 \pm 0.9$ from dense-grid, multi-seed evaluation (Section 5.4).

5.3.3 Multi-seed Experiments: AU and TwoNN are Stable, m^{knee} is Not

Table 4 reports MNIST VAE results with sparse grid $\mathcal{M} = \{4, 8, 12, 16, 20, 32, 64\}$ and three seeds (300 epochs).

Table 4: MNIST VAE multi-seed study: Exp-Mnist-Multi-300 (mean $\pm$ std, 3 seeds, sparse grid). AU and TwoNN ID are reproducible; m^{knee} varies across seeds.

m	AU (mean $\pm$ std)	TwoNN ID (mean $\pm$ std)	Note
4	4.0 $\pm$ 0.0	3.98 $\pm$ 0.14	all active
12	10.0 $\pm$ 0.0	7.41 $\pm$ 0.06	
32	15.7 $\pm$ 0.5	9.07 $\pm$ 0.10	
64	22.3 $\pm$ 0.5	10.03 $\pm$ 0.12	converges to $\hat{d}_{\text{ID}} \approx 10$
Sparse-grid m^{knee} ($\tau = 0.25$): seed 42=64, 123=32, 777=32; mean 43 $\pm$ 15			

Key findings. (1) AU values and TwoNN ID are highly reproducible across seeds ($\sigma \leq 0.5$ and $\sigma \leq 0.12$, respectively). (2) Sparse-grid m^{knee} varies from 32 to 64 ($\sigma = 15$), confirming that it is an unreliable diagnostic reference at coarse grids.

5.4 Dense-Grid Stabilisation of m^{knee}

To isolate the effect of grid density, we designed Exp-Mnist-Dense-300 with $\mathcal{M} = \{8, 9, 10, 11, 12, 13, 14, 15, 16, 18\}$ independent per- (m, seed) initialisation, 300 epochs, 3 seeds.

Table 5: Dense-grid vs. sparse-grid m^{knee} comparison (MNIST FC-VAE, $\beta = 4$). Grid density is the dominant stabilisation factor.

Condition	Max. grid in- terval	seed 42	seed 123	seed 777	Mean $\pm$ std
Sparse, 300ep (Exp-Mnist-Multi-300)	32	64	32	32	43 $\pm$ 15
Dense, 300ep (Exp-Mnist-Dense-300)	8	11	9	11	10.3 $\pm$ 0.9

The dense-grid result $m^{\text{knee}} = 10.3 \pm 0.9$ is consistent with TwoNN ID (≈ 10) and represents the main stabilised m^{knee} result of this paper. Note that even with dense grids the AU curve fluctuates non-monotonically (AU: 8 $\rightarrow$ 9 $\rightarrow$ 10 $\rightarrow$ 10 $\rightarrow$ 10 $\rightarrow$ 10 $\rightarrow$ 10 $\rightarrow$ 12 $\rightarrow$ 11 $\rightarrow$ 14 $\rightarrow$...), so m^{knee}

should still be treated as an exploratory heuristic, not a precise estimator. The fixed- τ method outperforms the second-difference method for dense grids ($\sigma = 0.9$ vs. 18.3 ± 6.8 , Appendix C).

5.5 TwoNN Stability: Bottleneck Margin Dominates

Three complementary experiments support Design Principle 1.

5.5.1 Swiss Roll: Isometry vs. Margin

For standard AE ($\kappa = 1.79$) and IsometricAE ($\kappa = 1.06$), TwoNN ≈ 1.55 at $m = 2 = d_{\text{true}}$ for both—the bottleneck constraint drives underestimation, not κ difference. At $m = 4 > d_{\text{true}}$, TwoNN recovers to ≈ 2.03 for standard AE despite $\kappa = 2.16$ (Appendix C, Table 9).

5.5.2 MNIST AE m -Sweep

TwoNN stabilises at $m \geq 48 \approx 5\hat{d}_{\text{ID}}$ (≈ 10.16) and remains stable even when κ jumps from 4.25 to 37.0 between $m = 64$ and $m = 128$ (TwoNN: $10.17 \rightarrow 10.98$, $\Delta < 0.82$).

5.5.3 Four-Model Comparison on MNIST/Fashion-MNIST (Exp-Real-Geom)

AE, IsometricAE, CAE, and DAE are swept over $m \in \{d, 2d, 3d, 5d, 10d\}$ ($d = \hat{d}_{\text{ID}} = 10$). All four models converge at $m = 5d$ – $10d$; inter-model TwoNN difference at the same m is < 0.3 , far smaller than the gain from increasing m (~ 2.5 points). These results strongly support Design Principle 1 (see Table 6 in main text and Appendix C for Fashion-MNIST).

Table 6: Exp-Real-Geom: MNIST four-model comparison (AE, IsometricAE, CAE, DAE) (mean $\pm$ std, 3 seeds; $\hat{d}_{\text{ID}} = 10$, representative m values). TwoNN stabilisation is driven by $m/\hat{d}_{\text{ID}}$ across all four models; inter-model difference at the same m is ≤ 0.3 .

Model	m	TwoNN ID	MLE ID	κ	Trust.	kNN ovlp.
AE	10 ($= d$)	7.3 ± 0.4	8.1 ± 0.3	85 ± 22	0.961	0.73
	30 ($= 3d$)	9.8 ± 0.2	10.5 ± 0.2	1200 ± 400	0.981	0.85
	100 ($= 10d$)	10.0 ± 0.1	10.8 ± 0.1	4.1×10^7	0.987	0.88
IsometricAE	10	7.5 ± 0.4	8.3 ± 0.3	3.2 ± 0.4	0.963	0.74
	30	9.9 ± 0.1	10.6 ± 0.1	5.2 ± 0.8	0.982	0.86
	100	10.1 ± 0.1	10.8 ± 0.1	7.8 ± 1.4	0.988	0.89
CAE	10	7.2 ± 0.5	8.0 ± 0.4	12.1 ± 3.2	0.958	0.72
	30	9.7 ± 0.2	10.4 ± 0.2	95 ± 31	0.980	0.84
	100	10.0 ± 0.1	10.7 ± 0.1	3200 ± 1100	0.986	0.87
DAE	10	7.6 ± 0.4	8.4 ± 0.4	18.5 ± 5.4	0.964	0.75
	30	9.9 ± 0.1	10.5 ± 0.2	183 ± 62	0.982	0.85
	100	10.0 ± 0.1	10.8 ± 0.1	6300 ± 2100	0.987	0.88

5.6 Fashion-MNIST: Preliminary External Check

Fashion-MNIST is used only as a preliminary external check, not as evidence of seed-level robustness. Single-seed (seed 42) results: TwoNN ID converges to ≈ 9.4 at $m \geq 32$, suggesting $\hat{d}_{\text{ID}} \approx 9$ – 10 (slightly lower than MNIST). VAE ($\beta = 4$) shows a sharp $\rho(m)$ drop around

$m = 12$ in this single run, but multi-seed reproducibility is not assessed. Fashion-MNIST 3-seed experiments are left for future work.

5.7 Conv-VAE: Applicability Boundary on MNIST

Conv-VAE (2-layer strided convolution encoder, 2-layer transposed convolution decoder) trained on MNIST (100 epochs) shows $AU \approx m$ up to $m = 64$ —no KL-induced saturation. Extending to 300 epochs with fixed β or cyclic KL annealing does not change this ($AU = m$ for $m \leq 64$; FC-VAE saturates at $AU = 15.7$ at $m = 32$ under the same conditions). Extending the m -grid to $\{96, 128, \dots, 256\}$ reveals $AU < m$ onset at $m \approx 96$, but no saturation plateau up to $m = 512$ (Appendix D).

High- β verification. Increasing $\beta_{\max}$ to 10 and 20 shifts the AU saturation to lower dimensions ($m^{\text{knee}} \approx 128$ and ≈ 96 , respectively). This is consistent with the water-filling interpretation: higher β raises $\beta\tau_{\text{eff}}^2$, activating fewer dimensions. The $\beta_{\max} = 4$ failure is therefore not a breakdown of the framework but a sign that $d^*(\beta) \gg 256$ under this setting, requiring stronger regularisation or decoder-capacity reduction.

5.8 Natural Images: Defining the Applicability Boundary

Table 7 summarises the key results.

Table 7: Applicability-boundary experiments on CIFAR-10 and SVHN. TwoNN is partially consistent near $m \approx 2\hat{d}_{\text{ID}}$; AU/m^{knee} is not stably identifiable under the tested conditions.

Dataset	Raw TwoNN	Latent TwoNN at $m = 64$	Latent TwoNN at $m = 256$	AU/m^{knee}
CIFAR-10	25.20	25.21 ± 0.41 (consistent)	30.91 (rising)	$AU = m$; boundary
SVHN	10.09	19.97 ± 0.12 (distorted)	27.08 (diverging)	$AU < m$ at $m \geq 128$; unreliable

Extended sweeps ($\beta_{\max}$ up to 80, thin decoder) yield plateau-like AU at $m \approx 512$ on CIFAR-10 (preliminary), but all stable-identification criteria remain unmet. Details in Appendix D.

TwoNN is consistent with the raw-data value near $m \approx 2\hat{d}_{\text{ID}}$ for CIFAR-10, but increases further at larger m (28.83 at $m = 128$, 30.91 at $m = 256$), so the consistency is not uniform across m . SVHN latent TwoNN diverges far from raw TwoNN (10.09 vs. 27.08 at $m = 256$), indicating strong geometric distortion in the latent space.

These experiments **quantify the applicability boundary**: AU/m^{knee} identification on natural images requires substantially stronger regularisation and decoder-capacity control than the tested settings provide.

6 Discussion

6.1 Role Separation is the Main Contribution

The core message of this paper is not that m^{knee} estimates effective dimensionality, but that *different indicators serve different diagnostic roles* and conflating them leads to misinterpreta-

tion. AU values are reliable monitors of KL-induced activity; TwoNN ID is a stable auxiliary reference (not an unbiased estimator); m^{knee} is informative only under controlled conditions.

6.2 When is m^{knee} Informative?

m^{knee} is informative as a diagnostic aid *only when*: (1) the m -grid is dense near $\hat{d}_{\text{ID}}$; (2) each (m, seed) is initialised independently; (3) at least 3 seeds are used; and (4) AU curves show reasonably monotone behaviour (observed in FC-VAE on MNIST). Under these conditions, $m^{\text{knee}} = 10.3 \pm 0.9$ aligns with TwoNN ID (≈ 10), providing a complementary check from the KL-dynamics side. Sparse grids ($m^{\text{knee}} = 43 \pm 15$) are unreliable and should be avoided.

6.3 Optimisation Dynamics Interpretation of m^{knee} Instability

Gradient competition hypothesis. The β -ELBO combines reconstruction gradient ($\nabla \mathcal{L}_{\text{rec}}$) and KL gradient ($\beta \nabla \mathcal{L}_{\text{KL}}$). Near $m \approx d^*(\beta)$, these gradients compete: the reconstruction term tries to activate surplus dimensions while the KL term tries to suppress them. The resolution speed depends on the initialisation, m -grid, and β , leading to a smooth (not step-like) AU curve in non-linear VAEs and thus grid-dependent m^{knee} values. This hypothesis is consistent with all observed results but has not been directly verified by gradient measurements; it should be regarded as an *interpretive hypothesis*.

6.4 Conv-VAE and Natural Images: Boundary, Not Failure

For Conv-VAEs and natural images, the inability to achieve stable m^{knee} identification is not a failure of the framework—it reveals the *applicability boundary*. The key insight from high- β experiments is that AU saturation does occur when regularisation is sufficiently strong relative to decoder capacity. The boundary is thus characterised by the effective τ^2 relative to the data eigenvalue spectrum, consistent with the water-filling interpretation of Analytical Result 1.

6.5 Limitations

- **Isometry bias of TwoNN ID.** The reference AE used in Stage 1 has $\kappa \approx 10^9$, so $\hat{d}_{\text{ID}}$ is an approximation. Numerical stability under $m \gg \hat{d}_{\text{ID}}$ is demonstrated, but geometric unbiasedness is not claimed.
- **Single-seed Fashion-MNIST.** Fashion-MNIST is used as a preliminary check only; multi-seed results are not available.
- **Linear-Gaussian approximation.** Analytical Result 1 is an analytical model for the linear-Gaussian surrogate; its applicability to non-linear VAEs is qualitative.
- **Computational cost.** Full m -grid sweeps (11 sizes $\times$ 300 epochs $\times$ 2 models) require hours to days on GPU. IsometricAE Jacobian computation scales as $O(m)$ and is impractical for large m .

- **Natural-image stable identification not achieved.** Under the tested conditions, stable identification on CIFAR-10/SVHN is not achieved; this defines the current applicability boundary.

7 Conclusions

We proposed a multi-indicator diagnostic framework for effective latent dimensionality in AE/VAE representations. The main contribution is not a new dimension estimator but a role separation: AU values as reliable activity monitors, TwoNN/MLE ID as numerically stable auxiliary references, and m^{knee} as an exploratory, grid-dependent change point.

Key findings: (1) AU values and TwoNN ID are reproducible across seeds (MNIST, 3-seed); (2) sparse-grid m^{knee} is unreliable ($\sigma = 15$), whereas dense-grid, independent-initialisation, multi-seed evaluation stabilises it at 10.3 ± 0.9 (TwoNN-consistent); (3) the dominant factor for TwoNN stability is bottleneck margin ($m \gg \hat{d}_{\text{ID}}$), not isometric regularisation; (4) Conv-VAE and natural-image experiments quantify the applicability boundary—AU/ m^{knee} diagnostics fail when decoder capacity greatly exceeds the regularisation level.

The proposed diagnostic protocol supports practitioners in avoiding over-interpretation of any single indicator, in designing m -grids that make m^{knee} informative, and in recognising when the framework reaches its limits.

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A Single-Seed VAE Illustration

Table 8 shows the single-seed preliminary run that motivated the multi-seed study. The $m^{\text{knee}} = 12$ value observed here depends on the specific grid ($m \in \{1, 4, 8, \dots, 256\}$) and initialisation, and is *not* used as a main result.

Table 8: MNIST VAE ($\beta = 4$, 300 epochs, seed 42, extended grid) — illustration only. The slowdown at $m = 12$ and the non-monotone AU (29 at $m = 64$, 26 at $m = 128$) reflect stochastic optimisation, not a stable signal. The main stabilised result is $m^{\text{knee}} = 10.3 \pm 0.9$ (Table 5).

m	MSE	AU	TwoNN ID	Note
1	0.0464	1	1.00	
8	0.0186	8	6.36	increasing
12	0.0184	8	6.26	ρ drops (single-seed, not main result)
64	0.0103	29	9.16	
128	0.0085	26	10.02	non-monotone AU
256	0.0072	36	10.14	

B Proofs and Supplementary Theory

B.1 Topological Lower Bound

Proposition 3 (Topological Lower Bound). *To represent a smooth d -dimensional manifold $\mathcal{M} \subset \mathbb{R}^N$ without self-intersections under zero reconstruction error (idealised setting), a bottleneck of $m \geq d_{\text{emb}}$ is necessary, where d_{emb} is the topological minimum embedding dimension. For simply connected manifolds (e.g. Swiss Roll), $d_{\text{emb}} = d_{\text{ID}}$. For the Torus T^2 , $d_{\text{ID}} = 2$ but*

$d_{\text{emb}} = 3$: it cannot be embedded in $\mathbb{R}^2$ without self-intersections, explaining the MSE elbow at $m = 3$ (see [11], Ch. 2).

Note: This is an idealised result; finite capacity and approximate reconstruction are excluded.

B.2 Proof Sketch for Analytical Result 1

The derivation follows the standard linear-Gaussian rate-distortion analysis with a β -ELBO objective. For each PCA direction j , the contribution $L_j = -D_j/(2\tau^2) - \beta R_j$ (rate-distortion surrogate) is maximised independently. Setting $\partial L_j / \partial D_j = 0$ yields the water-filling condition $D_j^* = \beta\tau^2$ (feasible only when $\lambda_j \geq \beta\tau^2$). Computing $\Delta L_j = L_j^{\text{active}} - L_j^{\text{dead}} = (\beta/2)h(\lambda_j/\beta\tau^2)$ where $h(t) = t - 1 - \log t \geq 0$ shows $\Delta L_j > 0$ iff $\lambda_j > \beta\tau^2$.

Remark 4 (KL vs. mutual information). *The surrogate L_j implicitly identifies the expected KL $\mathbb{E}_x KL(q_j \| p_j)$ with the mutual information $R_j = I(z_j; y_j)$. For a true VAE, the expected KL also includes aggregated-posterior/prior mismatch. Therefore Analytical Result 1 is a result for the linear-Gaussian rate-distortion surrogate, not a rigorous theorem about the full VAE ELBO.*

C TwoNN Stability Details and Linear Theory Bridge

C.1 Exp-Geom: MNIST AE m -sweep

Table 9: MNIST AE m -sweep (Exp-Geom). TwoNN stabilises at $m \geq 48 \approx 5\hat{d}_{\text{ID}}$.

m	κ	TwoNN ID	Trustworthiness	Convergence
4	2.69	4.21	0.960	not converged
16	3.28	8.17	0.993	converging
48	4.44	10.16	0.998	converged
64	4.25	10.17	0.998	stable
96	9.94	10.86	0.999	stable (despite κ jump)
128	37.0	10.98	0.999	stable
$n_{\text{train}} = 10\,000$, 200 ep, seed=42.				

C.2 Exp-Iso-ID: Standard AE vs. IsometricAE on Swiss Roll

Table 10: Latent TwoNN ID for AE and IsometricAE on Swiss Roll ($d_{\text{true}} = 2$, input TwoNN=2.427). At $m \geq 3$, TwoNN difference stays below $\pm 2.3\%$ despite κ ratio ~ 2 .

m	AE		IsometricAE		Diff. (%)
	TwoNN ID	κ	TwoNN ID	κ	
2	1.780	1.76	1.799	1.08	+1.1
3	2.458	2.10	2.454	1.11	−0.2
4	2.483	1.80	2.479	1.14	−0.2
6	2.421	1.79	2.474	1.10	+2.2
8	2.500	1.45	2.443	1.11	−2.3

C.3 Fashion-MNIST Exp-Real-Geom (Preliminary)

Fashion-MNIST ($\hat{d}_{\text{ID}} = 9$, $m \in \{9, 18, 27, 45, 90\}$) shows qualitatively the same pattern as MNIST: TwoNN converges at $m \approx 5\hat{d}_{\text{ID}}$, inter-model differences are small, and κ improvements have minimal effect. The Fashion-MNIST experiment is used as a preliminary single-seed external check; multi-seed validation is left for future work and is not used to support the main claims.

C.4 Linear Theory Bridge (Exp-Mnist-PCA)

MNIST PCA spectrum: $\lambda_1 = 5.22, \lambda_2 = 3.76, \dots, \lambda_{10} = 1.23$. At $\beta = 4$ and back-calculated $\tau_{\text{eff}}^2 \approx 0.30$, the water-filling prediction gives $d^* \approx 10$, consistent with TwoNN ID. The smooth (not step-like) AU curve of the non-linear FC-VAE explains why sparse-grid m^{knee} is unstable: gradual saturation violates the step-function assumption of Proposition 1.

D Conv-VAE and Natural-Image Details

D.1 Exp-Conv-M512: MNIST Conv-VAE Extended Grid

Table 11: MNIST Conv-VAE with $m \in \{256, 384, 512\}$ (per seed; KL cyclic annealing, $\beta_{\text{max}} = 4$, 150 ep). No AU plateau up to $m = 512$; $d^*(\beta) > 512$ under this setting.

m	AU (s42)	AU (s123)	AU (s777)
256	241	245	238
384	382	383	380
512	512	511	512

D.2 Natural-Image Extended Sweep (Exp-NatImg-Sweep)

Extended sweeps on CIFAR-10 ($m \in [32, 1024]$, $\beta_{\text{max}} \in \{10, 20, 40, 80\}$, Standard/Deep/Thin decoders, 300 ep, 3 seeds):

- $\beta_{\text{max}} = 80$, Thin decoder: AU plateau-like at $m \in [512, 1024]$ ($\text{AU} \approx 24$). This is a preliminary, conditional observation.
- TwoNN ID: 24.7–25.4 ($\sigma \leq 0.6$) across all conditions.

The five stable-identification criteria (seed stability, grid stability, β stability, AU-plateau stability, TwoNN–AU consistency) are all unmet under the tested conditions; strict stable identification remains a future goal.

E Sensitivity Analyses

E.1 AU Threshold δ

Swiss Roll and Torus experiments confirm AU patterns are identical for $\delta \in \{0.001, 0.01, 0.05, 0.1\}$ (active coordinate variance $O(1)$ vs. inactive $O(10^{-4})$; 2-order gap).

E.2 τ_{knee} Sensitivity

Cross-dataset validation confirms $\tau = 0.25$ is stable for Gaussian mixture data. Fashion-MNIST shows boundary sensitivity near $\tau \approx 0.25$; the second-difference alternative is recommended for gradual-slope curves. Dense-grid experiments confirm the fixed- τ method outperforms second differences ($\sigma = 0.9$ vs. 18.3 ± 6.8) for MNIST.

E.3 β Sensitivity

Preliminary 50-epoch MNIST subsets: $\beta = 1$ gives $\text{AU} \approx m$ (insufficient regularisation); $\beta = 8$ shows over-suppression signs. AU patterns are stable for $\beta \in [2, 8]$; $\beta = 4$ is used as default.